

All Four Forces from a Single Entanglement-Torque Mechanism: A Unified Sketch

Abstract

This document sketches how the same entanglement-torque mechanism that underlies our gravity and electromagnetism papers can be extended, with varying levels of rigor, to the weak and strong nuclear forces. The strongest empirical leg is the Bohr magneton, which is reproduced exactly at the electron scale with no G -dependent inputs. The gravity calculation reproduces the Mach–Sciama relation $G \sim c^2 R/M$ with predicted prefactor $4/5$; this is a consistency check rather than a free derivation of G , since the cosmic inputs are themselves G -dependent (see the gravity paper for the full circularity discussion). The weak and strong sectors are sketches: structurally parallel, but quantitatively dependent on inputs (W-boson mass, QCD scale) that are themselves measured rather than derived here. We are explicit throughout about which legs are quantitative predictions, which are structural matches, and which are organizational analogies.

1 STATUS OF THE FOUR LEGS

A blunt scorecard before getting into the structure:

- **Electromagnetism (clean).** The Bohr magneton $\mu_B = e\hbar/(2m_e)$ is reproduced exactly from the helical trajectory at the electron Compton scale, using only e , $\hbar$, m_e , and c . No G , no fitted parameters. This is the strongest empirical test of the mechanism.
- **Gravity (consistency at one scale, not local).** The mechanism produces a relation of the form $G = \gamma c^2 R/M$ with predicted $\gamma = 4/5$, consistent with the measured G when evaluated on cosmic inputs. But applied to any local causal patch (e.g. Earth: $M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.37 \times 10^6$ m), the same formula gives $G_{\text{eff}} \sim 10^{26}$ — off by 36 orders of magnitude. The relation therefore works only at the scale where M/R is forced by the critical-density condition, which itself contains G . The gravity calculation is a Mach–Sciama consistency check at one scale, not a local derivation. A truly local derivation would require following the Jacobson–Padmanabhan–Verlinde route (entropic gravity from local entanglement) and computing the entanglement bond density without using G as input — an open problem.
- **Weak force (sketch).** Structurally parallel via short-range jerk at the W mass scale; the coupling and range are taken as inputs, not derived.
- **Strong force (sketch).** Structurally parallel via confinement geometry at the QCD scale; again the coupling and scale are inputs.

Mass generation, charge quantization, and frame-dragging are discussed as organizational analogies, not as quantitative predictions.

2 UNIVERSAL MECHANISM

All four forces share the same underlying mechanism:

1. Every particle has an angular momentum budget $s = \sqrt{3}/2 \cdot \hbar$.
2. This budget is allocated across partners following the modular Hamiltonian structure $E = \hbar c/r$.
3. The Bisognano–Wichmann theorem shows the modular flow is a Lorentz boost, so the particle’s spin faces orthogonally to the displacement toward the partner.
4. The orthogonal jerk creates a helical trajectory with radius $\lambda_C = \hbar/(mc)$ and frequency $\omega = mc^2/\hbar$.
5. The force between two particles is $F = \alpha \cdot \hbar c/r^2$, where α is the coupling fraction (entanglement strength per pair).

The differences between forces arise from three parameters:

1. **Mass scale:** The particle mass determines the Compton wavelength and Compton frequency.
2. **Phase coherence:** Aligned phases (ferromagnetic) give coherent addition (N^2), random phases (thermal) give incoherent addition (N).
3. **Coupling structure:** Mass couples quadratically to the entanglement field ($\alpha \propto m^2$), charge couples linearly ($\alpha \propto e$).

3 GEOMETRIC FACTOR ACROSS FORCES

The geometric factor $\gamma = 4/5$ comes from the quadratic weight $\beta(r) = (1 - r/R)^2$ in the modular Hamiltonian. The Casini–Huerta–Myers theorem gives $\beta(r) = (R^2 - r^2)/(2R)$ for **any** spherical region in **any** d -dimensional CFT. This is theory-independent.

The Racah chain gives the quadratic weight when the filling fraction is low ($\rho \ll 1/2$). The cosmic baryon filling fraction is $\rho_b = 0.049$. The electron filling fraction is even lower:

$$\rho_e = \frac{\frac{4}{3}\pi\lambda_{C,e}^3}{V_{\text{universe}}/N_e} \approx 10^{-68} \quad (1)$$

Both are extremely low, so the quadratic-weight regime is the relevant one for both gravity and electromagnetism. We therefore expect the prefactor $\gamma = 4/5$ to apply across forces in their low-filling regime; we have not verified this for the weak and strong sectors, where filling fraction is not the controlling parameter.

4 PHASE COHERENCE AND THE FORCE RATIO

The ratio of electromagnetic to gravitational force between two protons is:

$$\frac{F_{\text{em}}}{F_{\text{grav}}} = \frac{\alpha_{\text{em}}}{\alpha_{\text{grav}}} = 1.24 \times 10^{36} \quad (2)$$

This decomposes into:

$$\frac{\alpha_{\text{em}}}{\alpha_{\text{grav}}} = \left(\frac{m_p}{m_e}\right)^2 \cdot \mathcal{C} = 3.40 \times 10^6 \cdot 1.84 \times 10^{29} \quad (3)$$

The phase coherence factor $\mathcal{C} \approx 1.8 \times 10^{29}$ encodes the fundamental difference between mass and charge coupling. Gravity's coupling fraction is **quadratic** in the mass ratio:

$$\alpha_{\text{grav}} = \left(\frac{\ell_P}{\lambda_{C,p}}\right)^2 \quad (4)$$

Electromagnetism's coupling fraction is **linear** in the charge-to-mass ratio:

$$\alpha_{\text{em}} = \frac{r_e}{\lambda_{C,e}} \quad (5)$$

where $r_e = e^2/(4\pi\epsilon_0 m_e c^2)$ is the classical electron radius. The quadratic versus linear scaling reflects the different ways mass and charge couple to the entanglement field. The coherence factor $\mathcal{C}$ is the residual from this difference in power-law scaling.

5 THE FINE STRUCTURE CONSTANT

The electromagnetic fine structure constant is:

$$\alpha_{\text{em}} = \frac{e^2}{4\pi\epsilon_0 \hbar c} = \frac{1}{137.036} \quad (6)$$

In the entanglement framework, this is the ratio of two length scales:

$$\alpha_{\text{em}} = \frac{r_e}{\lambda_{C,e}} = \frac{2.818 \times 10^{-15} \text{ m}}{2.426 \times 10^{-12} \text{ m}} = 7.297 \times 10^{-3} \quad (7)$$

The classical electron radius r_e is the length scale at which the electrostatic self-energy equals the rest mass:

$$\frac{e^2}{4\pi\epsilon_0 r_e} = m_e c^2 \quad (8)$$

The Compton wavelength $\lambda_{C,e}$ is the length scale at which the quantum bond energy equals the rest mass:

$$\frac{\hbar c}{\lambda_{C,e}} = m_e c^2 \quad (9)$$

The fine structure constant is the fraction of the quantum bond energy stored as electrostatic self-energy. It is the ratio of the electrostatic coupling to the quantum coupling at the electron scale.

6 WEAK AND STRONG NUCLEAR FORCES

The same mechanism applies at the W/Z boson mass scale (weak force) and the QCD scale (strong force).

6.1 Strong Force

The strong force operates at the QCD scale $\Lambda_{\text{QCD}} \sim 200 \text{ MeV}$:

$$\lambda_{\text{QCD}} = \frac{\hbar}{\Lambda_{\text{QCD}} c} = 1.0 \times 10^{-15} \text{ m} \quad (10)$$

The strong coupling at this scale is order unity:

$$\alpha_s \approx 1.0 \quad (11)$$

In the entanglement framework, $\alpha_s \approx 1$ means the entanglement is maximal. The confinement scale is where the entanglement energy equals the kinetic energy of quarks. At this scale, the coupling fraction is order unity, meaning the entanglement is saturated.

Asymptotic freedom (weak coupling at high energies, strong coupling at low energies) is the transition from the unresolved entanglement regime to the fully collapsed regime.

6.2 Weak Force

The weak force operates at the W boson mass scale ($M_W \approx 80.4 \text{ GeV}$):

$$\lambda_{\text{C},W} = \frac{\hbar}{M_W c} = 2.5 \times 10^{-18} \text{ m} \quad (12)$$

The weak coupling constant at this scale is:

$$\alpha_w \approx \frac{1}{30} \quad (13)$$

The short range of the weak force is due to the small Compton wavelength of the W/Z bosons. The Yukawa potential:

$$V(r) = -\frac{g^2}{4\pi r} e^{-M_W c r / \hbar} \quad (14)$$

has an exponential cutoff at $\lambda_{\text{C},W}$, the range of the weak force.

6.3 Summary of All Four Forces

Table 1: All four forces as entanglement torque at different mass scales.

Force	Coupling α	Mass scale	Length scale	Range
Gravity	5.9×10^{-39}	m_p	$1.32 \times 10^{-15} \text{ m}$	Infinite
Weak	3.3×10^{-2}	M_W	$2.5 \times 10^{-18} \text{ m}$	$\lambda_{\text{C},W}$
EM	7.3×10^{-3}	m_e	$2.43 \times 10^{-12} \text{ m}$	Infinite
Strong	~ 1.0	Λ_{QCD}	$1.0 \times 10^{-15} \text{ m}$	λ_{QCD}

7 MASS GENERATION (ORGANIZATIONAL SKETCH)

This section is also a sketch. We note that the standard Higgs result

$$m = \frac{yv}{\sqrt{2}} \quad (15)$$

admits an entanglement-style reading: the Yukawa coupling y can be viewed as a coupling fraction between the particle and the Higgs (vacuum) field, and the resulting mass as the entanglement energy stored in that interaction. This restates the Higgs mechanism in entanglement language; it does not derive the Yukawa values, which span six orders of magnitude (from $y_e \approx 3 \times 10^{-6}$ to $y_t \approx 1$) and remain a fundamental input.

What is not claimed: we are not deriving the mass hierarchy from first principles. We are observing that the entanglement framework is at least compatible with the Higgs picture.

8 CHARGE QUANTIZATION (ORGANIZATIONAL SKETCH)

This section is an organizational sketch, not a derivation. We note that the helical-trajectory picture is consistent with charge quantization in the same way the standard Dirac–Wilson argument is: the jerk-phase must return to its starting value after one Compton period, so the loop integral $\oint \mathbf{A} \cdot d\mathbf{l}$ must be a multiple of $2\pi\hbar/e$. This is the standard Dirac result

$$e \cdot g = \frac{n\hbar c}{2}, \quad (16)$$

restated in helical-trajectory language. We are not claiming an independent derivation of charge quantization beyond this restatement.

9 FRAME DRAGGING

Frame dragging is the precession of the local inertial frame due to the rotation of a mass. The Bisognano–Wichmann theorem identifies the modular Hamiltonian with the boost generator, so frame dragging is a boost effect in the rotating frame.

The GR prediction for Earth is:

$$\omega_{\text{FD}}^{\text{GR}} = \frac{2GJ}{c^2 R^3} = 1.9 \times 10^{-14} \text{ rad/s} \quad (17)$$

Our mechanism gives the per-nucleon precession:

$$\omega_{\text{pair}} = \frac{\alpha_{\text{grav}} \hbar c / R}{s} = 5.4 \times 10^{-37} \text{ rad/s} \quad (18)$$

A naive collective treatment (random walk, $\sqrt{N_{\text{Earth}}}$ nucleons) does not reproduce the GR prediction; the result is off by roughly a factor of 10^3 . The structural prediction (that frame dragging is a boost effect of modular flow under rotation) is matched, but the quantitative match is currently absent and is open work.

10 LIMITATIONS AND SCOPE

To close, a consolidated honest accounting of where the unified sketch stands:

- **Quantitative, G -free:** The Bohr magneton at the electron scale. This is the cleanest empirical leg.
- **Consistent at one scale, not local:** The cosmic relation $G = (4/5)c^2 R/M$. The prefactor $4/5$ is a genuine prediction, but the formula is not a local statement of G — it gives the right answer only where M/R is forced by the critical-density condition that itself contains G . Applied to any local causal patch the formula fails by tens of orders of magnitude. This is a Mach–Sciama consistency check, not a derivation of G .

- **Structural, quantitatively off:** Frame dragging. Correct structure (boost effect of modular flow under rotation), quantitative match still absent (off by $\sim 10^3$).
- **Order-of-magnitude:** Dark energy density. Within an order of magnitude of observation; precise factor requires holographic accounting beyond this scope.
- **Sketch only:** Weak and strong forces. Mass scales, couplings, and ranges are taken as inputs from the standard model, not derived. The framework is consistent with them, but does not predict them.
- **Organizational:** Mass generation, charge quantization. Restated in entanglement language; not derived from anything more fundamental.

The headline claim should therefore be: a single mechanism appears to underlie phenomena across all four forces, with one clean empirical test (the Bohr magneton) and a cosmological consistency check that reproduces a known structural relation with a predicted prefactor. The rest is sketch and analogy. We have tried throughout to be explicit about which is which.

11 OPEN PROBLEMS

1. **Yukawa couplings:** Mass hierarchy not derived.
2. **Fine structure constant value:** $\alpha_{\text{em}} = r_e/\lambda_{\text{C},e}$ is a relation, not yet a derivation of $1/137.036$ from deeper structure.
3. **Frame dragging collective factor:** Off by $\sim 10^3$; needs the right collective treatment of rotation-induced displacement.
4. **Strong-coupling running:** Asymptotic freedom not yet recovered from the entanglement picture.
5. **G -independent gravity test:** No experiment is currently proposed that would test the $4/5$ prefactor without using G -dependent cosmic inputs.
6. **Local derivation of G :** The most important open problem. Following the Jacobson–Padmanabhan–Verlinde program, derive Newtonian gravity from a local entanglement bond density η that is computable from QFT without G as an input. This would convert the cosmic relation from a one-scale consistency check into a derivation.